Summary & Closure Report

1. Project Information

- Project Name: Banking Loan Management System (BLMS)
- Testing Period:

Deliverables / Status	End Date	Start Date	Focus	Phase / Sprint
Executed Test Cases for Employee, Branch Customers & Accounts. Logged initial bugs.	1/9/2026	31/8/2026	Test Execution (User Stories 1-3)	Sprint 1
Verified fixes for Age/NID Triggers and Foreign Key constraints.	3/9/2026	2/9/2026	Bug Fixing & Retesting (Sprint 1)	Sprint 2
Executed Test Cases for Loans, Transactions, Automatic Fees & Collateral.	5/9/2026	4/9/2026	Test Execution (Remaining Stories)	Sprint 3
Retested Sprint 3 bugs, executed Full System Regression, and achieved 100% Pass Rate.	7/9/2026	6/9/2026	Bug Fixing & Final Regression	Sprint 4
Passed & Ready for Production	9/9/2026	8/9/2026	Final QA Summary Report	Sign-off

- 6 tester
- Bank
- My SQL

2. Test Scope Document: Bank & Loan Management System (BLMS2)

- **Project Overview & Objective**

The primary objective of this testing phase is to validate the backend business logic, relational data integrity, constraint enforcement, automated triggers, and transaction workflows of the **BLMS2** database schema across 4 planned sprints in Jira.

- **In-Scope (Functional & Database Testing)**

The following modules, business rules, and database automations are strictly included within the testing boundaries:

A. Customer Management & Automated Validations

- **Age Eligibility Verification:** Verification that customers under 21 years old are rejected during registration (date_of_birth validation).
- **National ID (NID) Expiry Enforcement:** Validation that National IDs with expired dates (nid_expiry_date <= CURDATE()) trigger system exceptions.
- **Automated Age Calculation:** Verification of the BEFORE INSERT trigger (trg_customers_before_insert) that dynamically computes and stores customer age upon insertion.
- **Uniqueness & Format Integrity:** Testing unique constraints on email, phone, and national_id.

B. Account & Branch Management

- **Account Type Coverage:** Functional testing across all supported account types (Savings, Current, Loan, Overdraft, Wallet).
- **Underage Account Block:** Verification of trg_accounts_before_insert to prevent account opening for customers under 21 years of age.
- **Branch Allocation:** Testing foreign key integrity connecting customers and employees to valid branches (branches table).

C. Loan & Collateral Management

- **Loan Application Rules:** Validation of iScore boundary checks (300 to 850), interest rate parameters, and valid loan types (auto_loans, personal_loans, students_loans, Mortgages).
- **Multi-Collateral Aggregation:** Testing one-to-many relationships between loans and collateral items (Property, Vehicle, Gold, Other).
- **Automated Collateral Roll-ups:** Verification of automated triggers (AFTER INSERT, AFTER UPDATE, AFTER DELETE on collaterals) that maintain:
    - total_collateral_amount: Sum of all collateral values tied to a specific loan.
    - collateral_details: Concatenated textual breakdown of each collateral item and its respective value.

D. Transaction Processing & Balance Automation

- **Transaction Execution:** Functional verification of deposit, withdrawal, loan Disbursement, and repayment transaction types.
- **Real-time Balance Updates:** Testing the database trigger (trg_transactions_after_insert) to ensure account balances update accurately:
    - Deposits & Loan Disbursements → Balance Increases
    - Withdrawals & Repayments → Balance Decreases

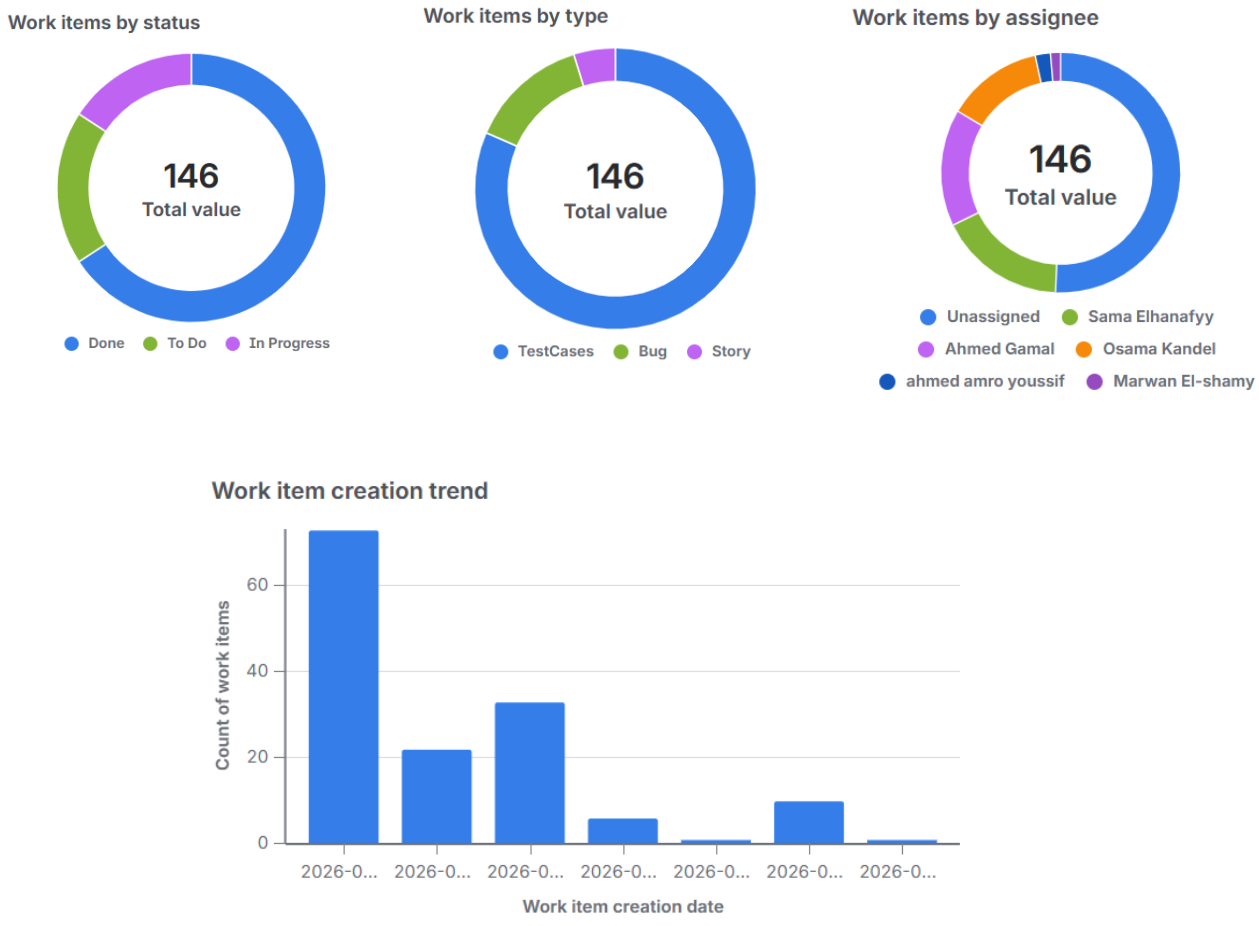
E. Automated Fee Engine (Central Bank Standard Compliance)

- **Rule-based Fee Calculation:** Verification of trg_calculate_transaction_fee executing dynamic fee percentages:
    - **Withdrawal Fee:** 2.00%
    - **Loan Administrative Fee:** 5.00%
    - **Repayment Processing Fee:** 1.00%
    - **Deposit Fee:** 0.00% (Free)

- **Audit Trail & Deduction:** Verification that calculated fees are logged directly into the fees table with associated transaction_id and automatically deducted from the customer's account balance.

F. Employee Management

- **Data Integrity:** Verification of employee registration details, branch linking, and National ID expiry check enforcement (trg_employees_before_insert).



Severity

Critical: 1

High: 3

Medium: 7

Low: 4

3. Major Findings

The business had a lot of problems.

4. Lessons Learned

Requirement analysis helped identify missing validation scenarios and uncovered gaps in the original requirements.

Based on the executed test cases and current defect status, the system is considered ready for UAT.